Proper Velocity Neural Networks

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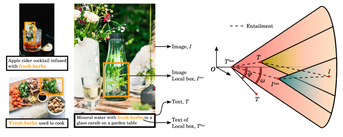
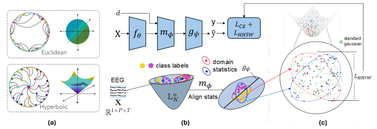
Motivation

Applications of Hyperbolic Spaces

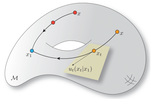
Vision-Language Models



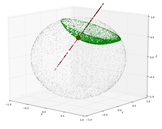
EEG



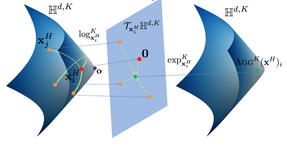
Generation



NLP



Graph



Popular Models

Operator	Poincaré ball $\mathbb{P}^n_K$	Hyperboloid $\mathbb{H}^n_K$
Definition	$\mathbb{P}^n_K = \{x \in \mathbb{R}^n \mid \|x\|^2 < -1/K\}$	$\mathbb{H}^n_K = \{x \in \mathbb{R}^{n+1} \mid \langle x,x\rangle_{\mathcal{L}} = 1/K, x_t > 0\}$
$g_x(w,v)$	$\left(\lambda^K_x\right)^2 \langle w,v\rangle, \quad \lambda^K_x = \frac{2}{1+K\|x\|^2}$	$\langle w,v\rangle_{\mathcal{L}} = \langle v_s, w_s\rangle - v_t w_t$
$\mathrm{d}(x,y)$	$\frac{2}{\sqrt{\lvert K\rvert}} \tanh^{-1}\left(\sqrt{\lvert K\rvert}\lVert -x \oplus_{\mathrm{M}} y\rVert\right)$	$\frac{1}{\sqrt{\lvert K\rvert}} \cosh^{-1}\left(K\langle x,y\rangle_{\mathcal{L}}\right)$
$\mathrm{Log}_x\, y$	$\frac{2}{\sqrt{\lvert K\rvert}\lambda^K_x} \tanh^{-1}\left(\sqrt{\lvert K\rvert}\lVert -x \oplus_{\mathrm{M}} y\rVert\right) \frac{-x\oplus_{\mathrm{M}} y}{\lVert -x\oplus_{\mathrm{M}} y\rVert}$	$\frac{\cosh^{-1}(\beta)}{\sqrt{\beta^2-1}}(y-\beta x), \quad \beta = K\langle x,y\rangle_{\mathcal{L}}$
$\mathrm{Exp}_x\, v$	$x \oplus_{\mathrm{M}} \left(\tanh\left(\sqrt{\lvert K\rvert}\frac{\lambda^K_x\lVert v\rVert}{2}\right)\frac{v}{\sqrt{\lvert K\rvert}\lVert v\rVert}\right)$	$\cosh(\alpha)x + \frac{\sinh(\alpha)}{\alpha}v, \quad \alpha = \sqrt{\lvert K\rvert}\lVert v\rVert_{\mathcal{L}}$
$\mathrm{PT}_{x\to y}(v)$	$\frac{\lambda^K_x}{\lambda^K_y}\mathrm{gyr}[y,-x]v$	$v - \frac{K\langle y,v\rangle_{\mathcal{L}}}{1+K\langle x,y\rangle_{\mathcal{L}}}(x+y)$

- Pros: closed-form operators
- Cons: **constrained** spaces

Proper Velocity Spaces

The PV model $\mathbb{PV}^n_K = \mathbb{R}^n$

$$g_x(u,v) = \langle u,v\rangle + K\beta_x^2\langle x,u\rangle\langle x,v\rangle, \quad \forall x \in \mathbb{PV}^n_K, \forall u,v \in T_x\mathbb{PV}^n_K. \qquad \beta_x = \frac{1}{\sqrt{1-K\|x\|^2}}$$

$$x \oplus_{\mathrm{U}} y = x + y + \left\{\frac{1-\beta_y}{\beta_y} - K\frac{\beta_x}{1+\beta_x}\langle x,y\rangle\right\}x,$$

$$t \otimes_{\mathrm{U}} y = \sinh\left(t\sinh^{-1}\left(\sqrt{-K}\|y\|\right)\right)\frac{y}{\sqrt{-K}\|y\|}, \quad (t\otimes_{\mathrm{U}} \mathbf{0} = \mathbf{0}).$$

- What is the expression of the **Riemannian operators**?
- What is the basic deep learning **layers**?

Riemannian Operators

Theorem 4.3 (PV Riemannian operators). [↓] *Let $\pi = \pi_{\mathbb{PV}^n_K \to \mathbb{P}^n_K}$. Given $x,y \in \mathbb{PV}^n_K$ and $v \in T_x\mathbb{PV}^n_K$, the Riemannian operators on the PV space are*

$$\mathrm{Exp}_x(v) = x \oplus_{\mathrm{U}} \left(\frac{1}{\sqrt{-K}}\sinh\left(\frac{\sqrt{-K}(1+\beta_x)}{\beta_x}\|d\pi_x(v)\|\right)\frac{d\pi_x(v)}{\|d\pi_x(v)\|}\right), \tag{10}$$

$$\mathrm{Log}_x(y) = \sigma(x,y)z + \tau(x,y)\langle x,z\rangle x, \tag{11}$$

$$\mathrm{PT}_{x\to y}(v) = \frac{1+\beta_x}{\beta_x}\tilde{v} - K\frac{(1+\beta_x)\beta_y}{(1+\beta_y)\beta_x}\langle y,\tilde{v}\rangle y, \tag{12}$$

$$\mathrm{d}(x,y) = \frac{2}{\sqrt{-K}}\tanh^{-1}\left(\sqrt{-K}\|\pi(-x\oplus_{\mathrm{U}} y)\|\right), \tag{13}$$

with $z = (-x)\oplus_{\mathrm{U}} y$. For the parallel transport, $\tilde{v} = \mathrm{gyr}_{\mathrm{M}}[\bar{y},-\bar{x}]\left(d\pi_x(v)\right)$ with $\mathrm{gyr}_{\mathrm{M}}$ as the Möbius gyration in App. B.4, $\bar{x} = \frac{\beta_x}{1+\beta_x}x$ and $\bar{y} = \frac{\beta_y}{1+\beta_y}y$. Here, the scalar coefficients in the logarithm are

$$\sigma(x,y) = \frac{2}{\sqrt{-K}}\frac{\tanh^{-1}\left(\sqrt{-K}\|\pi(z)\|\right)}{\|z\|}, \tau(x,y) = \frac{2\beta_x}{1+\beta_x}\frac{\sqrt{-K}\tanh^{-1}\left(\sqrt{-K}\|\pi(z)\|\right)}{\|z\|}.$$

At the identity $\mathbf{0}$, the above operators can be further simplified:

$$\mathrm{Exp}_{\mathbf{0}}(v) = \frac{1}{\sqrt{-K}}\sinh\left(\sqrt{-K}\|v\|\right)\frac{v}{\|v\|}, \qquad \mathrm{Log}_{\mathbf{0}}(y) = \frac{1}{\sqrt{-K}}\sinh^{-1}\left(\sqrt{-K}\|y\|\right)\frac{y}{\|y\|},$$

$$\mathrm{PT}_{\mathbf{0}\to y}(v) = v - K\frac{\beta_y}{1+\beta_y}\langle y,v\rangle y, \qquad \mathrm{PT}_{x\to\mathbf{0}}(v) = v + K\frac{\beta_x^2}{1+\beta_x}\langle x,v\rangle x,$$

$$\mathrm{d}(\mathbf{0},y) = \frac{1}{\sqrt{-K}}\sinh^{-1}\left(\sqrt{-K}\|y\|\right).$$

Theorem 4.4 (Gyro by Riemannian). [↓] *The PV gyro operations can be rewritten as*

$$x\oplus_{\mathrm{U}} y = \mathrm{Exp}_x\left(\mathrm{PT}_{\mathbf{0}\to x}(\mathrm{Log}_{\mathbf{0}}(y))\right), \quad t\otimes_{\mathrm{U}} x = \mathrm{Exp}_{\mathbf{0}}(t\,\mathrm{Log}_{\mathbf{0}}(x)) \quad \forall x,y \in \mathbb{PV}^n_K, \forall t \in \mathbb{R}.$$

Neural Network Layers

$$p(y=k \mid x) \propto \exp\left(v_k(x)\right),$$

MLR Classifier

Theorem 5.2 (PV MLR). [↓] *For $x \in \mathbb{PV}^n_K$, the score $v_k(x)$ in Eq. (18) for each class k is*

$$v_k(x) = \frac{\|z_k\|}{\sqrt{-K}}\sinh^{-1}\left(\cosh(\sqrt{-K}r_k)\frac{\sqrt{-K}}{\|z_k\|}\langle x,z_k\rangle - \sinh(\sqrt{-K}r_k)\sqrt{1-K\|x\|^2}\right), \tag{19}$$

where $z_k \in \mathbb{R}^n$ and $r_k \in \mathbb{R}$ are parameters for class k. In particular, as $K \to 0^-$ we have $v_k(x) \to \langle x,z_k\rangle + b_k$ with $b_k = -r_k\|z_k\|$, which recovers the Euclidean MLR in Eq. (14).

FC

Theorem 5.3 (PV FC layer). [↓] *The output $y = \mathcal{F}(x) \in \mathbb{PV}^m_K$ has the closed form*

$$y_k = \frac{1}{\sqrt{-K}}\sinh(\sqrt{-K}v_k(x)), \quad 1 \le k \le m, \tag{22}$$

where $v_k(x)$ is defined as Eq. (19) with $z_k \in \mathbb{R}^n$ and $r_k \in \mathbb{R}$ as the FC parameters. In particular, as $K \to 0^-$ we have $y_k \to \langle x,z_k\rangle + b_k$ with $b_k = -r_k\|z_k\|$, which recovers the Euclidean FC layer.

Convolution

$$y = \mathcal{F}\left(\mathrm{Concat}\left(x_i,\ldots,x_k\right)\right)$$

Activation

$$x \mapsto \mathrm{Exp}_{\mathbf{0}}\left(\sigma\left(\mathrm{Log}_{\mathbf{0}}(x)\right)\right) \quad \text{or} \quad x \mapsto \sigma(x)$$

Normalization

$$\forall i \le N, \quad \tilde{x}_i \leftarrow \overbrace{\beta\oplus_{\mathrm{U}}}^{\text{Biasing}}\left(\overbrace{\frac{s}{\sqrt{v^2+\epsilon}}\otimes_{\mathrm{U}}}^{\text{Scaling}}\left(\overbrace{-\mu\oplus_{\mathrm{U}} x_i}^{\text{Centering}}\right)\right),$$

$$\mu = \mathrm{FM}(\{x_i\}_{i=1}^N) = \mathrm{argmin}_{y\in\mathbb{PV}^n_K}\frac{1}{N}\sum_{i=1}^N \mathrm{d}^2(x_i,y), \quad v^2 = \frac{1}{N}\sum_{i=1}^N \mathrm{d}^2(x_i,\mu).$$

Experiments

Stability

Table 1: Failure and violation probabilities (%) of $r \otimes_{\mathcal{H}} x$ in FP32.

r	Failure rate			Violation rate		
	$\mathbb{PV}^n_K$	$\mathbb{P}^n_K$	$\mathbb{H}^n_K$	$\mathbb{PV}^n_K$	$\mathbb{P}^n_K$	$\mathbb{H}^n_K$
1	0	0	0	N/A	0	32.50
5	0	0	0	N/A	0	92.36
10	0	0	0	N/A	0	99.76
20	0	0	4.23	N/A	0	100
50	0	0	64.42	N/A	0	100
75	0	0	79.63	N/A	0	100
100	0	0	88.26	N/A	0	100
150	0	0	96.43	N/A	0	100
200	0	0	100	N/A	0	100
1000	0	0	100	N/A	0	100

Table 2: $\|\mathrm{Log}_{\mathbf{0}}(\mathrm{Exp}_{\mathbf{0}}(v)) - v\|$.

Model	FP32	FP64
$\mathbb{P}^n_K$	2.1×10^{-4}	4.3×10^{-11}
$\mathbb{H}^n_K$	1.0×10^{0}	1.0×10^{0}
$\mathbb{PV}^n_K$	2.1×10^{-7}	6.7×10^{-16}

Table 3: Gradient magnitude $\|\nabla_x\|$ across varying radii.

Model	$\|\nabla_x\|$ Range	Gradient behavior
$\mathbb{P}^n_K$	$[1.1 \times 10^{-11}, 7.6 \times 10^{-13}]$	Vanishing gradients
$\mathbb{H}^n_K$	[0, NaN]	Exploding gradients
$\mathbb{PV}^n_K$	$[1.1 \times 10^{-4}, 2.1 \times 10^{-6}]$	Stable gradients

Effectiveness

Image

Table 4: Accuracies of hyperbolic MLRs on ResNet-18. Best results are in **bold**. δ represents the δ-hyperbolicity (lower is more hyperbolic), which comes from Bdeir et al. (2024, Tab. 1).

Model	Method	CIFAR-10 ($\delta = 0.26$)	CIFAR-100 ($\delta = 0.23$)
$\mathbb{P}^n_K$	Poincaré MLR (Ganea et al., 2018, Eq. 25)	95.09 ± 1.51	76.78 ± 0.67
	Unidirectional MLR (Shimizu et al., 2021, Eq. 6)	95.12 ± 0.20	77.19 ± 0.10
$\mathbb{H}^n_K$	Lorentz MLR (Bdeir et al., 2024, Thm. 2)	95.02 ± 0.12	77.96 ± 0.09
$\mathbb{PV}^n_K$	PV MLR (with $\mathrm{Exp}_{\mathbf{0}}$)	95.27 ± 0.12	78.19 ± 0.59
	PV MLR (without $\mathrm{Exp}_{\mathbf{0}}$)	$\mathbf{95.30 \pm 0.18}$	$\mathbf{78.20 \pm 0.37}$

Graph

Table 5: Accuracies of hyperbolic networks on graph learning. Best results are in **bold**. δ represents the δ-hyperbolicity (lower is more hyperbolic).

Model	Method	Disease ($\delta = 0$)	Airport ($\delta = 1$)	PubMed ($\delta = 3.5$)	Cora ($\delta = 11$)
$\mathbb{K}^n_K$	KNN (Mao et al., 2024)	79.41 ± 0.55	92.10 ± 0.97	69.36 ± 0.76	52.26 ± 1.99
$\mathbb{P}^n_K$	HNN (Ganea et al., 2018)	79.90 ± 0.01	82.16 ± 2.95	69.28 ± 0.85	49.68 ± 1.25
	HNN++ (Shimizu et al., 2021)	80.57 ± 0.23	88.40 ± 0.17	73.68 ± 0.39	52.06 ± 0.90
$\mathbb{H}^n_K$	LNN (Bdeir et al., 2024)	79.90 ± 0.01	75.20 ± 1.08	68.82 ± 0.88	$\mathbf{53.34 \pm 1.65}$
$\mathbb{PV}^n_K$	PVNN	$\mathbf{81.15 \pm 0.23}$	$\mathbf{97.96 \pm 0.42}$	$\mathbf{74.33 \pm 0.22}$	51.42 ± 1.33

Genome

Table 10: Comparison (MCC) of hyperbolic and Euclidean convolutional networks on TEB datasets.

Task	Dataset	Euclidean CNN	HCNN-S	PVCNN
Retrotransposons	LINEs	70.63 ± 1.24	76.12 ± 2.16	$\mathbf{81.83 \pm 0.27}$
	SINEs	85.15 ± 1.64	85.45 ± 1.16	$\mathbf{93.78 \pm 0.54}$
DNA transposons	hAT-Ac	87.45 ± 0.90	89.61 ± 1.34	$\mathbf{92.08 \pm 0.80}$
Pseudogenes	processed	60.66 ± 0.82	68.30 ± 0.93	$\mathbf{71.27 \pm 0.78}$
	unprocessed	51.94 ± 2.69	56.10 ± 0.56	$\mathbf{62.31 \pm 0.78}$

Ablations

Table 6: Results of Tangent FC (TFC) vs PV FC, and Tangent BN (TBN) vs GyroBN.

Method	Disease	Airport	PubMed	Cora
PVNN+TFC	80.86 ± 0.30	86.99 ± 0.61	$\mathbf{74.40 \pm 0.43}$	$\mathbf{53.58 \pm 0.81}$
PVNN	$\mathbf{81.24 \pm 0.36}$	$\mathbf{97.93 \pm 0.29}$	74.16 ± 0.32	52.26 ± 1.32
PVNN+TBN	80.67 ± 0.38	98.71 ± 0.36	73.52 ± 0.12	45.36 ± 2.44
PVNN+GyroBN	$\mathbf{81.24 \pm 0.19}$	$\mathbf{99.03 \pm 0.18}$	$\mathbf{74.34 \pm 0.31}$	$\mathbf{46.64 \pm 5.45}$

Table 9: Ablations on PV activations.

Method	Disease	Airport	PubMed	Cora
Tangent Act.	81.24 ± 0.36	97.93 ± 0.29	74.16 ± 0.32	$\mathbf{52.26 \pm 1.32}$
FC σ	81.34 ± 0.43	$\mathbf{99.40 \pm 0.15}$	74.02 ± 0.17	51.34 ± 0.46
FC σ + Tangent Act.	80.96 ± 0.19	99.15 ± 0.38	73.96 ± 0.22	51.30 ± 1.65
Euc. Act.	$\mathbf{81.34 \pm 0.43}$	98.87 ± 0.35	$\mathbf{74.56 \pm 0.59}$	38.10 ± 3.30